

Technical Document #350: Deep Learning - Comprehensive Overview

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Transformer architecture relies entirely on self-attention mechanisms, dispensing with recurrence and convolutions. It has become the foundation for modern NLP models.

Long short-term memory (LSTM) networks are a type of RNN that can learn long-term dependencies. They use gating mechanisms to control the flow of information.

Transfer learning is a technique where a model trained on one task is re-purposed on a second related task. It has revolutionized computer vision and NLP.

Attention mechanisms allow models to focus on different parts of the input when producing output. They have become essential in sequence-to-sequence models.

Recurrent neural networks (RNNs) are designed to process sequential data. They maintain hidden states that capture information about previous elements in the sequence.

Batch normalization normalizes the inputs of each layer, reducing internal covariate shift. It allows higher learning rates and acts as a regularizer.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Key Takeaways:

- Residual connections enable training of very deep neural networks by providing shortcuts for gradient flow.
- Batch normalization normalizes the inputs of each layer, reducing internal covariate shift.
- Long short-term memory (LSTM) networks are a type of RNN that can learn long-term dependencies.